

Chart-Aware Local Association Structure on the L^∞ Atlas

A model object, a bridge from conditional measures of association,
and an experimental program for vaginal 16S and metagenomic data

Design brief

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Abstract

The LPS/PS-LPS program equips every sample with a chart (a local PCA parametrization) over a neighbor graph, and fits the local *conditional mean* of features and outcomes on that chart. The conditional measures of association developed in the `no5` manuscript (functional divergences δ^0, δ^1 , the monotonicity coefficient, order-0 and order-1 conditional correlations, the signature correlation, the co-change correlation, and correlation typing) summarize how those mean surfaces co-move. This brief argues that, for boundary-supported compositional data such as the vaginal microbiome, the biologically interesting object — microbes interacting with each other and with the host immune system — is *not* captured by mean co-movement, and is also not addressable by interior-only models (Aitchison log-ratios, logistic-normal, Poisson-log-normal), because those models place zero probability on the boundary where the data actually live. We build the association layer on the L^∞ atlas of [1] instead, where the max-reference chart is zero-safe and subcompositionally coherent. The resulting *chart-aware local association object* factorizes into two layers that the existing measures cannot see: a **co-occurrence layer** (a discrete graphical model on presence indicators, living on the face lattice of the L^∞ cube) and a **conditional-abundance layer** (a Gaussian/nonparanormal graphical model on max-referenced log-ratios of the present taxa). We make both layers precise, give weighted local estimators and a PS-LPS-style synchronized field estimator (Algorithms 1–2), prove that the `no5` suite factors through the mean field and is therefore blind to both new layers (Section 5), and lay out a falsifiable experimental program (Section 6) wired to the `gflow/geosmooth` tooling and to the France et al. (2020) VALENCIA 16S data ($n \approx 13,231$) and the ZAPPS/PreSSMat metagenomic data ($n \approx 2,600$).

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1 Setup, notation, and the object we want

Data. We observe n samples. For sample j , let $x_j = (x_{j0}, \dots, x_{jd}) \in \mathbb{R}_{\geq 0}^{d+1}$ be a vector of nonnegative abundances over $p = d + 1$ features (ASVs, genes, or metabolites), and let $h_j \in \mathbb{R}^q$ be host/clinical covariates (e.g. pH, Nugent score, gestational age, cytokine panel, pregnancy status, outcome labels). Compositional data carry information only up to scale, so a sample is a point of the nonnegative real projective space $\mathbb{RP}_{\geq 0}^d = \{[x] : x \in \mathbb{R}_{\geq 0}^{d+1} \setminus \{0\}\}$, where $[x]$ is the ray spanned by x . Write $s_j = \mathbf{1}\{x_j > 0\} \in \{0, 1\}^p$ for the *presence pattern* (applied coordinatewise) and $A_j = \text{supp}(x_j) = \{i : x_{ji} > 0\}$ for the *active set*.

The three-layer stack. The current pipeline is a tower:

- (i) a *graph* $G = (V, E)$ on samples (adaptive-radius / intersection kNN) that answers “who is my neighbor”;
- (ii) a *chart* at each anchor (local PCA giving Euclidean coordinates) on which LPS/PS-LPS fits the local conditional mean of a response or feature;

- (iii) an *association* read-out (the **no5** measures) summarizing how fitted mean surfaces co-move locally.

The premise of this brief is that the omics snapshot is a projection of a deeper generative process — a microbial community and a host immune system in mutual regulation — so that the natural object sitting on each chart is not a single fitted mean but a *local conditional joint law* $P(x, h \mid \text{chart coordinates near } x)$, or at least its dependence structure. Every **no5** measure is a scalar or matrix functional (a compression) of that law. The goal of Sections 2–4 is to name the richer object and make it estimable; Section 5 recovers the existing measures as functionals of it.

Why interior models are the wrong substrate. The closure/normalization of compositional data introduces a sign artifact in raw correlations, and the standard fixes assume strictly positive support.

Proposition 1.1 (Closure sign bias). *Let $P = (P_0, \dots, P_d)$ be a random composition.*

- (a) *If $X \sim \text{Multinomial}(N, p)$ then for $i \neq k$, $\text{Cov}(X_i, X_k) = -Np_i p_k < 0$ and $\text{Corr}(X_i, X_k) = -\sqrt{\frac{p_i p_k}{(1-p_i)(1-p_k)}} < 0$.*
- (b) *If $P \sim \text{Dirichlet}(\alpha)$ with $\alpha_0 = \sum_m \alpha_m$ then $\text{Corr}(P_i, P_k) = -\sqrt{\frac{\alpha_i \alpha_k}{(\alpha_0 - \alpha_i)(\alpha_0 - \alpha_k)}} < 0$ for all $i \neq k$.*

Hence every pair is negatively associated by construction, and the Dirichlet covariance is a rank-deficient, sign-locked function of the mean: it cannot represent a co-occurring guild.

The interior-only repair (additive/centered log-ratio, logistic-normal, and the count-level Poisson-log-normal) moves the dependence into an unconstrained latent space where it can take either sign, but it assumes $x_{ji} > 0$ for all i, j . For the human vaginal microbiome this is biologically false: most taxa are genuinely absent from most women (a *structural zero*), not merely undersampled. Poisson-log-normal in particular models every zero as a sampling zero of a strictly positive latent intensity, fabricating a fully connected latent network. The presence pattern s_j is itself the dominant signal (it is essentially what a community state type encodes), so the substrate must treat the boundary — and the support — as first-class.

Notation summary. φ_k is the L^∞ chart of cell k ; $\xi = \varphi_k(x) \in [0, 1]^d$ its cube coordinate; $u_i = \log(x_i/x_{\max}) \leq 0$ the max-referenced log-ratio; $w_c(\cdot)$ kernel weights at chart c ; π presence probabilities, B co-occurrence (Ising) edges, μ abundance mean, Ω abundance precision; $\mathcal{A}(x) = (\pi, B, \mu, \Omega)(x)$ the local association object; $\Theta(x)$ the LPS conditional-mean field that the **no5** measures read.

2 The L^∞ atlas as the modeling substrate

We recall the L^∞ parametrization [1] and add the two pieces the association layer needs: a *soft* partition of unity (so charts overlap, as PS-LPS requires) and the face stratification (so the support is explicit).

Definition 2.1 (L^∞ normalization and cells). The L^∞ normalization $\pi_\infty : \mathbb{R}_{\geq 0}^{d+1} \setminus \{0\} \rightarrow \Delta_\infty^d$, $\pi_\infty(x) = x/\|x\|_\infty$ with $\|x\|_\infty = \max_i x_i$, is the single-component (max) ratio transform. The L^∞ simplex decomposes into $d+1$ top cells $\Delta_\infty^d = \bigcup_{k=0}^d Q_k$, where $Q_k = \{x \in \Delta_\infty^d : x_k = 1\} = \{x : x_k \geq$

$x_i \forall i$ is the set of samples whose dominant (argmax) component is k . Each Q_k is homeomorphic to the cube $[0, 1]^d$ via the chart

$$\varphi_k([x]) = \left(\frac{x_i}{x_k} \right)_{i \neq k} \in [0, 1]^d, \quad x_k = \max_i x_i.$$

Proposition 2.2 (Zero-safety and coherence). *On Q_k the chart φ_k is well defined (the denominator x_k is the maximum, hence strictly positive for $x \neq 0$) and extends continuously to the boundary strata where other coordinates vanish; a structural zero $x_i = 0$ maps to $\xi_i = 0$, an interior point of the cube’s parametrization rather than a singularity. Moreover φ_k is subcompositionally coherent with respect to deletion of non-maximal (low-abundance) components [1]: deleting components $i \notin \{k\}$ commutes with coordinate projection.*

This is exactly the property the log-ratio charts lack: the additive log-ratio $\phi_0(x) = (x_i/x_0)_{i \neq 0}$ blows up when the fixed reference $x_0 = 0$, and the centered log-ratio is not subcompositionally coherent (Example 3-B of [1]). The max reference is adaptive and never zero.

Construction 2.3 (Soft L^∞ atlas / partition of unity). Hard cell assignment by the argmax is unstable for near-tie samples (two taxa almost co-dominant), which are precisely the community-state transitions where dysbiosis lives. Replace the hard indicator by overlapping chart weights. For inverse temperature $\beta > 0$ define

$$\omega_k(x) = \frac{(x_k/\|x\|_\infty)^\beta}{\sum_m (x_m/\|x\|_\infty)^\beta}, \quad \sum_k \omega_k(x) = 1.$$

As $\beta \rightarrow \infty$, $\omega_k(x) \rightarrow \mathbf{1}\{k = \arg \max_i x_i\}$ and we recover the hard L^∞ cells. For finite β , a sample contributes to every chart whose reference is near-maximal, with weight decaying in the gap to the dominant component. We call $\{(Q_k, \varphi_k, \omega_k)\}_k$ the soft L^∞ atlas. Truncation/merging of low-occupancy cells into L^∞ -CSTs [1] is applied first, so k ranges over the retained CSTs.

Definition 2.4 (Face stratification). Within a cell, the cube $[0, 1]^d$ is stratified by which coordinates are zero: for an active set $A \ni k$, the open face $F_A = \{\xi : \xi_i > 0 \ (i \in A \setminus k), \ \xi_i = 0 \ (i \notin A)\}$ collects samples with presence pattern A . The association object will be defined *conditional on the face* F_A : a discrete law over which face a sample occupies, and, given the face, a continuous law on its interior.

Remark 2.5 (Local charts compose with the atlas). Inside a cell, the LPS chart (data-adaptive local PCA) is applied to the cube coordinates ξ , or to the present log-ratios $u = (u_i)_{i \in A \setminus k}$, not to raw abundances. Thus the LPS/PS-LPS machinery runs unchanged on L^∞ -chart coordinates; `chart.dim (auto/local.auto)` controls the local dimension of *both* the mean model and, below, the abundance covariance model.

3 The model object: a two-layer local association law

Fix a chart (anchor) and, conditionally on the soft atlas, work in its cube / log-ratio coordinates. The central claim is that for boundary-supported data the local dependence structure *factorizes* into a combinatorial support layer and a continuous abundance layer, and that these are biologically distinct objects (guild membership vs. within-guild interaction) that must be modeled separately.

3.1 Layer A — co-occurrence (support) structure

The presence pattern $s = \mathbf{1}\{x > 0\}$ records which face of the cube a sample occupies. Model it as a conditional discrete graphical model (an Ising / pairwise log-linear model), localized to cell k and modulated by host covariates h .

Definition 3.1 (Co-occurrence layer). Conditional on cell k and covariates h , the presence vector $s \in \{0, 1\}^p$ follows

$$P(s \mid k, h) \propto \exp\left(\sum_i a_i(k, h) s_i + \sum_{i < j} B_{ij}(k) s_i s_j\right),$$

with node potentials $a_i(k, h) = a_i^0(k) + \langle c_i(k), h \rangle$ (host covariates shift colonization log-odds) and edge potentials $B_{ij}(k)$. The *co-occurrence graph* of cell k has an edge $\{i, j\}$ iff $B_{ij}(k) \neq 0$, equivalently $s_i \perp s_j \mid s_{\setminus\{i,j\}}, h$ fails: two taxa are conditionally co-present (mutually attracting, $B_{ij} > 0$) or mutually exclusive ($B_{ij} < 0$) given the rest of the community and the host state.

This is the layer that is structurally invisible to interior models (no zeros, so $s \equiv \mathbf{1}$) and, as Section 5 shows, to the mean-based `no5` measures. For the vaginal microbiome it should encode *Lactobacillus* mutual exclusion and the *Gardnerella–Atopobium–Sneathia–Prevotella* anaerobe consortium.

3.2 Layer B — conditional-abundance dependence

Conditional on the active set $A = \text{supp}(x) \ni k$, the present taxa carry continuous information through the max-referenced log-ratios $u_i = \log(x_i/x_k) \leq 0$, $i \in A \setminus \{k\}$, which are zero-safe by conditioning (every x_i , $i \in A$, is strictly positive). Let $\mu_A(x) = \mathbb{E}[u \mid \text{chart at } x]$ be the local conditional mean (the LPS surface) and $r = u - \mu_A(x)$ the residual.

Definition 3.2 (Conditional-abundance layer). On face F_A , assume a nonparanormal model: there exist monotone marginal transforms $g = (g_i)$ such that $g(r) \sim \mathcal{N}(0, \Sigma_A(x))$, with precision $\Omega_A(x) = \Sigma_A(x)^{-1}$. The *conditional-abundance graph* has an edge $\{i, j\}$ iff $\Omega_{A,ij}(x) \neq 0$ ($u_i \perp u_j \mid u_{\setminus\{i,j\}}$ fails), and the signed local interaction strength is the partial correlation

$$\rho_{ij}(x) = -\Omega_{A,ij}(x) / \sqrt{\Omega_{A,ii}(x) \Omega_{A,jj}(x)}.$$

Setting $g = \text{id}$ recovers a Gaussian graphical model on log-ratios (the L^∞ -chart analogue of SPIEC-EASI [10], but on the boundary-safe max reference rather than the `clr`); the nonparanormal g relaxes Gaussianity and is the natural continuation of the monotonicity-first viewpoint of `no5`.

Remark 3.3 (Two orthogonal kinds of association). Layer A is about *whether* taxa appear together; Layer B is about *how*, among those present, abundances co-fluctuate beyond what their mean surfaces explain. A microbe–microbe “interaction” in the biological sense — conditional dependence given community context — is an edge of A (colonization dependence) and/or of B (abundance regulation), never a property of the mean surfaces alone.

3.3 The chart-aware association field and host coupling

Definition 3.4 (Local association object and field). At location x define the *local association object*

$$\mathcal{A}(x) = (\pi(x), B(x), \mu(x), \Omega(x)),$$

Require: anchor x_c , cell k , soft weights ω_k , bandwidth h_{band} , presence threshold τ , penalties λ_A, λ_B .

- 1: $w_{c,j} \leftarrow \omega_k(x_j) K(d_G(x_c, x_j)/h_{\text{band}})$ for all j (*neighborhood weights*)
- 2: **Layer A:** for each i , weighted ℓ_1 -logistic regression of s_i on $(s_{\cdot, \setminus i}, h)$ with weights $w_{c,\cdot}$; collect into $\widehat{B}_c$, symmetrize (AND-rule); set $\widehat{\pi}_{c,i}$ to the weighted fitted presence probability at x_c .
- 3: $A_c \leftarrow \{i : \text{weighted presence fraction} \geq \tau\}$ (*active set*)
- 4: $u_{ji} \leftarrow \log(x_{ji} / \max_m x_{jm})$ for $i \in A_c \setminus \{k\}$.
- 5: $\widehat{\mu}_c \leftarrow \text{FIT.LPS}(u \mid L^\infty\text{-chart}; \text{chart.dim}, \text{design.basis})$; $r_{ji} \leftarrow u_{ji} - \widehat{\mu}_{c,i}(x_j)$.
- 6: (optional) nonparanormal transform $r \leftarrow \widehat{g}(r)$.
- 7: $\widehat{\Omega}_c \leftarrow \text{WGLASSO}(r, w_{c,\cdot}, \lambda_B)$; $\widehat{\rho}_{c,ij} \leftarrow -\widehat{\Omega}_{c,ij} / \sqrt{\widehat{\Omega}_{c,ii} \widehat{\Omega}_{c,jj}}$.

return $\mathcal{A}_c = (\widehat{\pi}_c, \widehat{B}_c, \widehat{\mu}_c, \widehat{\Omega}_c)$.

Algorithm 1: LOCALTWO_LAYERFIT — association object at one chart

collecting presence probabilities $\pi_i(x) = P(s_i = 1 \mid x)$, co-occurrence edges $B(x)$, abundance mean $\mu(x)$, and abundance precision $\Omega(x)$, all defined as soft-atlas, neighborhood-weighted quantities at x . The assignment $x \mapsto \mathcal{A}(x)$ is the *chart-aware association field*; it is piecewise-defined over the L^∞ atlas and glued across cell boundaries by Construction 2.3.

Host/immune variables enter both layers, giving a single mixed graphical model over heterogeneous nodes (Bernoulli s_i , Gaussian u_i , and h) in the exponential-family MRF sense [13]. Its blocks are interpretable: microbe–microbe edges (within A and within B), microbe–host edges (host modulating presence in A, host modulating abundance coupling in B), and host–host edges. This is the precise object behind “microbes and the host immune system interacting”: it is now a field of graphs over the community-state space, and an edge may appear in one L^∞ -CST and vanish in another.

4 Estimation and PS-LPS-style synchronization

4.1 Local weighted estimator

At a chart c with anchor x_c and cell $k(c)$, define neighborhood weights combining the soft-atlas membership and a graph kernel,

$$w_{c,j} = \omega_{k(c)}(x_j) K(d_G(x_c, x_j)/h_{\text{band}}),$$

with d_G the graph distance and K a kernel. Layer A is fit by weighted ℓ_1 -penalized nodewise logistic regression (neighborhood selection [8]); Layer B by a weighted graphical lasso [7] on the LPS residuals of the present log-ratios (optionally after a nonparanormal marginal transform [9]).

4.2 Synchronized field — the PS-LPS analogue for second moments

PS-LPS synchronizes neighboring charts’ *predictions* (first moments) via a prediction-overlap penalty. The association field admits the identical move on *second moments* and on the support layer: tie overlapping charts’ parameters together, and tie adjacent cells along their shared boundary faces.

Require: charts $\{\mathbf{c}\}$, overlaps ζ , penalties $\lambda_A, \lambda_B, \gamma, \eta_A, \eta_B$.

1: warm start: $\mathcal{A}_{\mathbf{c}} \leftarrow \text{LOCALTWO LAYERFIT}(\mathbf{c})$ for all $\mathbf{c}$ (Algorithm 1), in parallel.

2: **repeat** (*block coordinate / ADMM*), for each chart $\mathbf{c}$:

(a) update $(\mu_{\mathbf{c}}, \Omega_{\mathbf{c}})$: weighted graphical-lasso step with a proximal pull toward $\sum_{\mathbf{c}'} \zeta_{\mathbf{c}\mathbf{c}'}(\mu_{\mathbf{c}'}, \Omega_{\mathbf{c}'});$

(b) update $(B_{\mathbf{c}}, a_{\mathbf{c}})$: weighted nodewise-logistic step with a proximal pull toward $\sum_{\mathbf{c}'} \zeta_{\mathbf{c}\mathbf{c}'} B_{\mathbf{c}'};$

until the change in $\{\mathcal{A}_{\mathbf{c}}\}$ is below tolerance.

return synchronized field $\{\mathcal{A}_{\mathbf{c}}\}.$

Algorithm 2: SYNCHRONIZEDFIELD — fused estimation over the atlas

Definition 4.1 (Synchronized field objective). Let $\zeta_{\mathbf{c}\mathbf{c}'} \geq 0$ measure chart overlap (shared weighted samples, or $\sum_j \min(w_{\mathbf{c},j}, w_{\mathbf{c}',j})$). Estimate $\{\mathcal{A}_{\mathbf{c}}\}$ by minimizing

$$\sum_{\mathbf{c}} \left[\ell_{\mathbf{c}}^A(B_{\mathbf{c}}, a_{\mathbf{c}}) + \ell_{\mathbf{c}}^B(\mu_{\mathbf{c}}, \Omega_{\mathbf{c}}) + \lambda_A \|B_{\mathbf{c}}\|_1 + \lambda_B \|\Omega_{\mathbf{c}}\|_1 \right] + \gamma \sum_{\mathbf{c} \sim \mathbf{c}'} \zeta_{\mathbf{c}\mathbf{c}'} D_{\mathbf{c}\mathbf{c}'}(\mathcal{A}_{\mathbf{c}}, \mathcal{A}_{\mathbf{c}'}),$$

where ℓ^A, ℓ^B are the weighted (pseudo-)log-likelihoods of Layers A and B, and the fusion term, restricted to coordinates shared by the two charts, is

$$D_{\mathbf{c}\mathbf{c}'} = \underbrace{\|\mu_{\mathbf{c}} - \mu_{\mathbf{c}'}\|_2^2}_{\text{PS-LPS term}} + \eta_A \|B_{\mathbf{c}} - B_{\mathbf{c}'}\|_F^2 + \eta_B \|\Omega_{\mathbf{c}} - \Omega_{\mathbf{c}'}\|_F^2.$$

The first term is exactly the PS-LPS prediction-overlap penalty; the other two are its lifts to the co-occurrence and precision layers. Across a cell boundary, $\mathbf{c}$ and $\mathbf{c}'$ are charts of different cells sharing a near-tie face, so $D_{\mathbf{c}\mathbf{c}'}$ enforces continuity of the association field through community-state transitions.

4.3 Uncertainty, identifiability, and cost

Uncertainty. Reuse the `no5` Bayesian-bootstrap + permutation machinery: resample Dirichlet sample weights to obtain posterior draws of each edge (B_{ij}, ρ_{ij}) and of the presence probabilities, and permute host labels for null distributions; report edges by posterior sign-stability and by the Wasserstein significance measure already used for δ^1 . Stability selection [14] over subsamples gives per-edge selection frequencies.

Identifiability and cost. A full $p \times p$ precision per chart is infeasible, but is never required: Layer B lives on the active set $A_{\mathbf{c}}$, which is small (typical vaginal communities have a handful of co-present dominant taxa), so $\Omega_{\mathbf{c}}$ is low-dimensional; the fusion term borrows strength across charts; and `chart.dim` bounds the working dimension. Layer A is p -dimensional but sparse and is fit by p independent weighted logistic regressions (embarrassingly parallel). The main statistical risk is rare-taxon edges in Layer A (few presence events): regularize hard and report selection frequency rather than point edges.

5 The bridge: no5 measures as read-outs of the field

We now make precise the claim that the existing conditional measures of association are a (lossy) compression of $\mathcal{A}$. Let $\Theta(x) = (\Theta_i(x))_i$ be the *mean field*: the vector of fitted conditional

expectations $\Theta_i(x) = \widehat{\mathbb{E}}[\cdot_i \mid X = x]$ (the LPS surfaces $\widehat{y}_G, \widehat{z}_G$). Note Θ is exactly the μ -block of $\mathcal{A}$; the support law π, B and the residual precision Ω are the remaining blocks.

Proposition 5.1 (The **no5** suite factors through the mean field). *Each of the functional divergences δ^0, δ^1 , the monotonicity coefficient mono , the order-0 and order-1 conditional correlations $\text{corr}_X^0, \text{corr}_X^1$, the signature correlation $\text{corr}_{\text{sign}}^1$, the co-change correlation Δcorr , and the correlation-typing map is a functional of Θ and $\nabla\Theta$ alone. Consequently, if two association fields $\mathcal{A}, \mathcal{A}'$ share the same mean field Θ but differ in (π, B, Ω) , every measure in the suite returns identical values.*

Proof sketch. By inspection of the **no5** definitions: $\delta^k(Y\|Z)$ and $\Delta^k(Y\|Z)$ integrate $|d^k E(Y|Z)/dZ^k|$, functionals of a single mean surface; mono is their ratio. $\text{corr}_X^0(Y, Z)$ correlates the standardized conditional means $E(Y|X), E(Z|X)$; corr_X^1 takes the cosine of $\nabla E(Y|X), \nabla E(Z|X)$; the signature correlation reads the sign pattern of those gradients; Δcorr averages sign of products of increments of $\widehat{y}_G, \widehat{z}_G$. All reference only $\Theta = \widehat{E}(\cdot|X)$ and its gradient/differences; none references the residual covariance Σ (hence Ω) or the presence law (hence π, B). The claim follows. $\square$

Proposition 5.2 (Co-change is sign-concordance of the mean field). *For fitted means,*

$$\Delta\text{corr}_G(\widehat{y}_G, \widehat{z}_G \mid x) = \frac{1}{|U_x|} \sum_{x' \in U_x} \text{sign}[(\Theta_y(x') - \Theta_y(x))(\Theta_z(x') - \Theta_z(x))],$$

which depends on Θ only through the signs of its local increments; it is invariant to any strictly monotone reparametrization of Θ and to (π, B, Ω) . In the chain-graph (1D) limit it equals $\text{sign}(\dot{\Theta}_y \dot{\Theta}_z)$, the order-1 conditional correlation. Thus co-change is a discrete sign-concordance — a coarsened Kendall functional — of the mean field.

Remark 5.3 (What the suite cannot see, made operational). Proposition 5.1 says the two new layers live in the kernel of the **no5** suite. Two falsifiable contrasts isolate them:

- (a) *Co-occurrence invisibility.* Two taxa that are mutually exclusive (never co-present) have a strong negative Layer-A edge $B_{ij} \ll 0$, yet their co-change is undefined/ vanishing because they are rarely simultaneously present to difference. Layer A detects them; the suite does not.
- (b) *Residual-interaction invisibility.* Two taxa whose mean surfaces are parallel ($\nabla\Theta_i \parallel \nabla\Theta_j$, so $\text{corr}^1 \approx 1$, co-change ≈ 1) but whose residual log-ratios are conditionally independent have $\Omega_{ij} = 0$: strong mean coupling, no interaction. The suite reports near-perfect association; Layer B reports none.

These power Experiment 6.

Signature correlation, precisely. The **no5** signature correlation classifies the mean-gradient vector $\nabla\Theta$ by which face of the L_1 sphere it occupies — it is the combinatorial type of the *mean* coupling. Its natural companion at Layer B is the sign-and-support pattern of $\Omega(x)$ (the partial-correlation graph), and at Layer A the sign pattern of $B(x)$ (attraction vs. exclusion). The existing construction supplies the first; the model object adds the other two, giving a full combinatorial fingerprint ($\text{sign } \nabla\Theta, \text{sign } B, \text{sign } \Omega$)(x) at each location.

Correlation typing, upgraded. The **no5** typing map $x \mapsto [\text{corr}^{0/1}(z_i, z_j \mid x)]_{ij} \in \text{Sym}^p(\mathbb{R})$ is, by Proposition 5.1, a Θ -level matrix field. Replace it by the *augmented typing*

$$\mathcal{T}(x) = (B(x), \Omega(x)) \in \{\text{signed graphs}\} \times \text{Sym}^{|A|}(\mathbb{R}),$$

Table 1: Existing `gflow/no5` constructs as functionals of the association field, the layer they see, and how they are repurposed for the model object. Θ : mean field; $\nabla\Theta$: mean gradient; Ω : abundance precision (Layer B); B, π : co-occurrence (Layer A).

Construct	Functional computed	Layer seen	Repurposing
$\delta^0, \delta^1, \text{mono}$	shape/monotonicity of Θ_i	Θ	per-feature mean diag
corr_X^0	corr of conditional means	Θ	mean-coupling block
corr_X^1	cosine of $\nabla\Theta_i, \nabla\Theta_j$	$\nabla\Theta$	mean-coupling block
$\text{corr}_{\text{sign}}^1$	L_1 -face of $\nabla\Theta$	$\text{sign } \nabla\Theta$	one third of the finger
$\Delta\text{corr}(\text{lcor sign})$	sign-concordance of increments	$\text{sign } \nabla\Theta$	feed residuals $\rightarrow$ Layer
$\text{lcor}(\text{derivative})$	edge-difference correlation	$\Theta/\nabla\Theta$	on residuals $r \Rightarrow \hat{\rho}_{ij}$
$\text{lslope}, \text{lslope.neighborhood}$	asymmetric $\nabla\Theta$ slope	$\nabla\Theta$	directed mean coupling
$\text{gfcor}, \text{gfassoc}.*$	basin-level association	Θ over basins	cell-level summaries of
pearson.wcor	weighted correlation	—	inner-product engine
$\text{lcor.with.posterior}, \text{permutation.test.lcor}$	BB / null	—	uncertainty for B, ρ
correlation typing	$x \mapsto [\text{corr}]_{ij}$	Θ	augment to $\mathcal{T} = (B, \Omega)$

optionally with the Θ block retained. Because the original typing is a function of the mean block alone, $\mathcal{T}$ strictly refines it; clustering the image $\mathcal{T}(G)$ yields metabo-types / CST-subtypes defined by *co-occurrence and residual interaction*, not merely by mean co-movement, and these can then be tested against clinical outcomes exactly as in `no5`.

Repurposing the existing primitives. Crucially, the same `gflow` backends estimate the new layers if fed the right inputs: the symmetric edge-difference engine behind `lcor` computes Layer-B quantities when applied to residual log-ratios r (Algorithm 1, line 7) instead of fitted means, and `pearson.wcor` is the weighted inner product engine for the precision step.

6 Experimental program

Each experiment targets one piece of the structure above and is stated with a *falsification criterion*: a concrete result that would kill or seriously wound the corresponding claim. Two datasets are used: the combined vaginal 16S rRNA data of France et al. (2020) [5] ($n \approx 13,231$ samples, 199 taxa, VALENCIA CST labels available) and the ZAPPS/PreSSMat metagenomic data ($n \approx 2,600$, VIRGO 2 gene catalog). Tooling is named explicitly: `geosmooth::fit.lps/fit.ps.lps`; `gflow::lcor`, `lslope`, `lslope.neighborhood`, `gfcor`, `pearson.wcor`, `permutation.test.lcor`, `lcor.with.posterior`; graph construction via `dgraphs`; and, for the new layers, the standard `glasso/huge/JGL` estimators plus `SpiecEasi` and `SparCC` as interior-model baselines.

6.1 E1 — the atlas substrate

Experiment 1 (L^∞ decomposition vs. VALENCIA CSTs). **Claim.** The L^∞ atlas is a zero-safe, sample-independent substrate whose cells reproduce the established CST structure (the premise of Sections 2).

Procedure. Compute L^∞ -cell assignment $k(j) = \arg \max_i x_{ji}$ for all 13,231 samples; build the $n_0=50$ truncated decomposition (L^∞ -CSTs); cross-tabulate against VALENCIA CST/sub-CST labels; report adjusted Rand index and the cell $\times$ CST contingency table. Sweep n_0 and the sub-cube refinement depth; record cell occupancy (expect ≈ 13 cells covering $\geq 96\%$ of samples, top two $\approx 60\%$).

Expected. High concordance ($L. iners \rightarrow \text{III}$, $L. crispatus \rightarrow \text{I}$, $Gardnerella/Atopobium/Sneathia \rightarrow \text{IV-B}$, $\text{BVAB1} \rightarrow \text{IV-A}$, $L. jensenii \rightarrow \text{V}$), matching Tables in [1].

Falsification. If L^∞ -CSTs do not align with VALENCIA (ARI low) or occupancy is not concentrated, the atlas is not a faithful substrate and the stratification must be rethought before building layers on it.

6.2 E2 — the co-occurrence layer

Experiment 2 (Layer A recovers guilds; closure correlation does not). **Claim.** Definition 3.1 captures co-presence structure invisible to closure-biased correlation and to mean-based measures.

Procedure. Within each major L^∞ -CST, fit Layer A by weighted nodewise ℓ_1 -logistic regression of presence indicators s (neighborhood-selected Ising), with stability selection over 200 subsamples. Compare the resulting signed co-occurrence graph to (i) the empirical Pearson correlation of closed relative abundances and (ii) Layer A without covariates.

Expected. Positive edges within the anaerobe consortium (*Gardnerella–Atopobium–Sneathia–Prevotella*) and negative edges between *Lactobacillus* spp. and anaerobes; the closure correlation (Proposition 1.1) shows the spurious all-negative pattern, missing positive guild structure.

Falsification. If Layer A edges are unstable (selection frequency near chance) or coincide with the closure-correlation sign pattern everywhere, the co-occurrence layer adds nothing beyond the artifact.

6.3 E3 — the conditional-abundance layer

Experiment 3 (Max-reference precision vs. pseudocount log-ratio). **Claim.** Layer B on the boundary-safe max reference avoids the pseudocount distortion that interior log-ratio methods incur near the boundary.

Procedure. Within *L. crispatus* and *L. iners* L^∞ -CSTs, estimate Ω on residual max-referenced log-ratios (Algorithm 1: `fit.lps` residuals $\rightarrow$ weighted `glasso`); compare partial-correlation networks to `clr+pseudocount glasso` / `SpiecEasi` and to `SparCC`. Sweep the pseudocount $\epsilon \in \{10^{-6}, \dots, 1\}$ and quantify edge turnover (the sensitivity analysis [1] flags as routinely neglected). Use `pearson.wcor` for the weighted second moments and `lcor.with.posterior` for edge BB intervals.

Expected. Max-reference networks stable; `clr+pseudocount` networks swing materially with ϵ , especially for taxa with many structural zeros.

Falsification. If max-reference and `clr` networks agree and neither depends on ϵ , the boundary problem is empirically negligible for these data and the interior repair suffices.

6.4 E4 — boundary reorganization and dysbiosis

Experiment 4 (The field reorganizes across CST transitions). **Claim.** The association field changes across L^∞ -cell boundaries, and those boundaries are the dysbiosis transitions (Construction 2.3).

Procedure. Identify near-tie samples ($x_{(1)}/x_{(2)} < 1 + \epsilon$) on the *L. crispatus* $\leftrightarrow$ *Gardnerella* interface; estimate $\mathcal{A}$ on each side and along the soft overlap; track edge appearance/disappearance and partial-correlation sign flips as a function of the dominance gap. Overlay a clinical outcome (Nugent/Amsel BV, or sPTB where available) and run the `no5` gradient-flow / `gfcor` machinery on the outcome to confirm flow concentrates at the interface.

Expected. Distinct interaction graphs on the *Lactobacillus* vs. anaerobe sides, with reorganization (not mere rescaling) through the transition, co-located with elevated BV risk.

Falsification. If $\mathcal{A}$ is constant across the boundary, the “field of graphs” framing is unnecessary — a single global network would do.

6.5 E5 — synchronization

Experiment 5 (Fused field vs. independent per-chart fits). **Claim.** PS-LPS-style synchronization (Algorithm 2) improves the association layer as prediction synchronization improved the mean.

Procedure. Compare independent LOCALTWO_LAYERFIT against the fused estimator on held-out log-likelihood (Layer B), edge selection stability (Layer A), and parameter-field smoothness; sweep γ, η_A, η_B . Mirror the PS-LPS screened-vs-full-grid protocol and the failure/telemetry accounting used in `geosmooth` runs.

Expected. Fusion lowers held-out deviance and raises edge stability in low-occupancy charts, with the largest gains near boundaries.

Falsification. If fusion never beats independent fits on held-out likelihood, second-moment synchronization is not justified (even if first-moment PS-LPS was).

6.6 E6 — the bridge, falsified directly

Experiment 6 (The suite is blind to Layers A and B). **Claim.** Propositions 5.1–5.2: the `no5` measures factor through Θ and miss (π, B, Ω) .

Procedure. (a) Mutual-exclusion contrast: pick taxa pairs with strong Layer-A negative edges (E2) and show their `lcor/lslope` co-change is null/undefined. (b) Parallel-mean contrast: pick pairs with $\text{corr}^1 \approx 1$ (`lcor type="derivative"`) and show $\hat{\rho}_{ij} \approx 0$ after residualization. (c) Typing comparison: cluster the image of the Θ -level typing vs. the augmented $\mathcal{T} = (B, \Omega)$ typing; compare cluster-CST and cluster-outcome association.

Expected. Clear cases where the suite and the layers disagree; augmented typing associates more strongly with CST/outcome.

Falsification. If no such pairs exist and augmented typing never beats Θ -typing, the new layers carry no independent signal in these data.

6.7 E7 — host/immune coupling

Experiment 7 (Covariate-modulated presence and abundance). **Claim.** Host variables enter both layers (Section 3.3).

Procedure. With available metadata (pH, Nugent, gestational age, pregnancy, and cytokines where measured), add h to Layer A node potentials and Layer B means; test covariate-modulated colonization (does low pH / pregnancy shift anaerobe presence odds?) and covariate-modulated abundance coupling (does the *Gardnerella-Atopobium* partial correlation depend on host state?), with permutation nulls via `permutation.test.lcor`-style label shuffles.

Expected. Significant microbe-host edges concentrated in the IV-B/IV-A L^∞ -CSTs.

Falsification. If host covariates never shift presence odds or coupling, the mixed-graph coupling is unsupported and microbe/host layers separate.

6.8 E8 — metagenomic transfer and cross-modal typing

Experiment 8 (Gene-level atlas and 16S $\leftrightarrow$ metagenome typing). **Claim.** The construction transfers to gene-level metagenomics and supports cross-modal correlation typing (the $X_{\text{TX}}/X_{\text{MB}}$ idea of `no5`).

Procedure. Repeat E1–E3 on the $n \approx 2,600$ VIRGO 2 data; test whether gene-level L^∞ -cells (e.g. *Limosilactobacillus oris*, *L. crispatus* genes) refine the 16S strata; compute the augmented typing $\mathcal{T}$ of metagenomic interactions conditional on the 16S L^∞ -CST and vice versa, and cluster into metabo-/geno-types.

Expected. Coherent gene-level cells nested within taxonomic CSTs; cross-modal types that stratify outcome.

Falsification. If gene-level cells are incoherent or cross-modal typing yields no stable clusters, the multi-omics extension does not hold at this depth.

7 Practical notes, difficulties, and first sprint

Within-cell support heterogeneity. A shared dominant taxon does not imply a shared present-set; samples in one L^∞ -CST still occupy many faces. Layer A carries this heterogeneity, so it needs adequate per-cell sample size; this is the main reason to run the truncated decomposition first and to fuse across charts (Algorithm 2).

Near-tie instability and the bandwidth β . Hard argmax assignment is unstable when $x_{(1)} \approx x_{(2)}$. The soft atlas (Construction 2.3) regularizes this, but β is a genuine tuning parameter: too large reproduces hard-cell instability, too small over-smooths across biologically distinct CSTs. Select β by held-out association-layer likelihood (it is the analogue of the LPS bandwidth).

Parameter counts. Layer B is only as large as the active face, which is small; Layer A is p -wide but sparse and parallelizable. The binding constraint is rare-taxon presence events in Layer A — regularize hard, lean on stability selection, and never report a Layer-A edge supported by a handful of co-presences.

Zero handling is the whole point. Validate that the pipeline never introduces a pseudocount in the max-reference path (Layer B conditions on presence, so its log-ratios are finite by construction). E3 is partly a guard against accidentally reintroducing the interior assumption.

Gaussian vs. nonparanormal. Start Gaussian (plain `glasso` on residual log-ratios) for speed; switch to the nonparanormal transform if marginal residuals are visibly non-Gaussian. The non-paranormal also keeps the layer continuous with the monotonicity-first philosophy of `no5`.

Minimal first sprint. The core thesis can be validated or killed with three runs on the 16S data, in order:

- S1. E1** — confirm the substrate (cheap; mostly reuses the L^∞ decomposition and VALENCIA labels). If E1 fails, stop.
- S2. E2 + E6(a)** — fit Layer A in the two largest L^∞ -CSTs and show a concrete mutual-exclusion edge that `lcor`/co-change cannot see. This is the cheapest decisive test of the central “compression” claim.
- S3. E3** — fit Layer B with the max reference and run the pseudocount sensitivity sweep against `SpiecEasi`. This is the cleanest test that the boundary geometry matters in practice.

If S1–S3 land, proceed to synchronization (E5), boundaries/outcome (E4), host coupling (E7), and the metagenomic transfer (E8).

Table 2: Claim $\rightarrow$ experiment $\rightarrow$ kill criterion.

Claim	Exp.	Kills it if ...
L^∞ atlas is a faithful, zero-safe substrate	1	L^∞ -CSTs do not track VALENCIA; occupancy not concentrated
Co-occurrence layer carries real guild signal	2	edges unstable or identical to closure-artifact signs
Boundary-safe precision $\neq$ pseudo-count precision	3	networks agree and are ϵ -insensitive
Field reorganizes across CST transitions	4	$\mathcal{A}$ constant across the boundary
Second-moment synchronization helps	5	fusion never beats independent fits out-of-sample
no5 suite is blind to A and B	6	no disagreeing pairs; augmented typing no better
Host enters both layers	7	covariates never shift presence odds or coupling
Construction transfers to metagenomics	8	gene cells incoherent; no stable cross-modal types
Soft-face / effective-dimension beats hard support	9	refined strata never beat exact-zero faces out-of-sample; ξ -vs- u and ℓ_1 -vs- ℓ_2 diagnostics flat on synthetic truth

8 Addendum: refining the face stratification with local intrinsic dimension

The face stratification of Definition 2.4 partitions each cell by *which coordinates are exactly zero*: the face F_A collects samples whose active set is $A = \text{supp}(x)$, and it has nominal dimension $|A| - 1$. This is a *combinatorial* stratification — read directly off the presence pattern $s = \mathbf{1}\{x > 0\}$ — and it is the scaffold on which both layers sit. This addendum argues that the combinatorial face is only an *upper bound* on the geometry the model should actually use; that the gap is exactly what local-PCA dimension estimation (`chart.dim`, in `auto/local.auto` mode) is built to find; and that closing the gap repairs three things the main text leaves slightly open — the working dimension of Layer B, the (currently unsoftened) presence threshold, and the choice of fusion penalty for the support layer.

In plain terms. The face tells you *which* microbes are present in a sample. But “present” is not the same as “doing something.” A taxon can be present yet pinned just above the detection limit, never really moving; and a handful of present taxa can move so tightly together that they trace out a curve rather than fill a region. So the count of present microbes (the face dimension) can badly overstate the number of *independent directions the community actually explores* near a sample. That smaller number is the local intrinsic dimension, and it is what the charts and the covariance model should be built on.

8.1 Two notions of dimension, currently conflated

The remark after Definition 2.4 and the identifiability note of Section 4.3 let the active-set size $|A| - 1$ (equivalently `chart.dim`) set the working dimension of Layer B: “ Ω_c is low-dimensional ... `chart.dim` bounds the working dimension.” But $|A| - 1$ is the *combinatorial cap*; the *statistical*

effective dimension $k \leq |A| - 1$, the number of directions in which the present taxa actually co-vary, is generally smaller. When $k < |A| - 1$, the $(|A| - 1) \times (|A| - 1)$ precision Ω_A is near-singular with $|A| - 1 - k$ collapsed directions, and the weighted graphical lasso is being asked to estimate a model larger than the data support. The local-PCA dimension is precisely the *effective rank* of that covariance.

Definition 8.1 (Effective local dimension and refined stratum). Fix a chart $\mathbf{c}$ with anchor $x_{\mathbf{c}}$ in cell k and active set $A = \text{supp}(x_{\mathbf{c}})$. Let $\Sigma_{\mathbf{c}}(h)$ be the soft-atlas, neighborhood-weighted covariance of the *cube* coordinates $\xi_A = (\xi_i)_{i \in A \setminus k}$ at bandwidth h , with eigenvalues $\lambda_1 \geq \dots \geq \lambda_{|A|-1} \geq 0$. The *effective local dimension* $k_{\mathbf{c}}(h)$ is the complexity-penalized minimizer of the local reconstruction cost

$$c_{\mathbf{c}}(r; h) = \sum_{m > r} \lambda_m(h) + \text{pen}(r), \quad 1 \leq r \leq |A| - 1,$$

(equivalently, the rank before a scale-relative spectral gap). The *refined stratum* at $x_{\mathbf{c}}$ is the pair $(A, k_{\mathbf{c}})$: the combinatorial face A gives the ambient (boundary) dimension, $k_{\mathbf{c}}$ the dimension the community actually fills, $1 \leq k_{\mathbf{c}} \leq |A| - 1$.

Consequence for Layer B: estimate Ω on the $k_{\mathbf{c}}$ retained directions (or ridge the $|A| - 1 - k_{\mathbf{c}}$ collapsed ones explicitly), rather than on the full active set. This is the principled answer to the “parameter counts” difficulty of Section 7: the binding dimension is $k_{\mathbf{c}}$, not $|A| - 1$.

In plain terms. Imagine three present taxa whose relative abundances always satisfy one fixed trade-off (when one rises another falls in lockstep). There are three taxa, but only *two* of them move freely — the third is determined. Fitting a full three-way interaction model invents structure that is not there. The effective dimension counts the freely-moving directions, and the covariance model should use that count.

8.2 The soft face: local dimension as the boundary analogue of β

Construction 2.3 carefully *softens* the dominant-taxon (cell) assignment with the temperature β , precisely because a hard argmax is unstable at the near-ties “where dysbiosis lives.” But the *face* assignment $A = \{i : x_i > 0\}$ is a hard threshold on the minor coordinates — and it is left unsoftened. With real 16S detection limits, “ $x_i > 0$ ” (one read versus zero) is noise, so the face boundary carries exactly the near-tie instability the design diagnosed for cells, now at the presence/absence edge. The fix is symmetric to β : a scale-aware effective active set.

Definition 8.2 (Soft face at scale h). Replace the hard active set by the *effective active set* $A_h(x) = \{i : \bar{\xi}_i(x) > \tau_h\} \cup \{k\}$, where $\bar{\xi}_i$ is the neighborhood-weighted cube coordinate and τ_h a scale-coupled threshold: a taxon counts toward the face only if it is present *and* sits above the resolution h . As $h \rightarrow 0$, $A_h \rightarrow \text{supp}(x)$ (the hard face); for finite h , taxa pinned within τ_h of the boundary are folded into the lower stratum. Select τ_h (equivalently the soft-face scale), as β is selected, by held-out association-layer likelihood.

Two reductions are now distinguished. The *soft face* (Def. 8.2) drops near-zero taxa — an *axis-aligned* reduction toward a lower simplex face, where the thin direction is a *named* coordinate. The *effective dimension* (Def. 8.1) additionally drops collapsed *oblique* directions among the retained taxa (the trade-off above), which are not aligned with any face. The axis-aligned case is the easier one, and the simplex helps: because you know *which* coordinate is heading to zero, you can run a *directed* variance-collapse test on that one coordinate, far more powerful than estimating the whole spectrum from a small neighborhood.

In plain terms. β answers “which taxon is in charge here?” softly, so that a sample near a 50/50 tie contributes to both charts. The soft face answers “which taxa really count here?” softly, so that a taxon barely above zero is not forced to be a full extra dimension. Both replace a brittle hard switch with a smooth, tunable one; the design built the first and skipped the second.

8.3 There is no single “true” dimension: tie the scale to the bandwidth

Effective dimension is inherently *scale-dependent*. Points sampled from a thin tube of radius ρ around a k -dimensional sub-face read as k -dimensional at any resolution coarser than ρ (the thin directions sit below the floor) and as $|A| - 1$ -dimensional at resolutions finer than ρ . There is no scale-free answer; there is a dimension-versus-scale profile. The clean resolution for this program: **set the dimension scale equal to the LPS/association bandwidth** h_{band} , because the chart only needs to be faithful at the scale the local model actually uses. Estimate $c_c(r; h)$ across a small band of scales and trust a dimension that is *persistent* across scales; a dimension that drops only at large h is a coarse-graining artifact, not structure. (This is the local, supervised-by-bandwidth analogue of multi-scale intrinsic-dimension and singularity detection; cf. [20, 21].)

In plain terms. Zoom out and a thin ribbon looks like a line; zoom in and you see it has width. Asking “what dimension is it?” has no answer until you say how closely you are looking. So look at exactly the scale the smoother smooths at, and only believe a dimension that survives across a range of zoom levels.

8.4 The coordinate trap: estimate dimension in ξ , model dependence in u

Layer B lives in the max-referenced log-ratios $u_i = \log(x_i/x_k) \leq 0$. As a taxon heads toward a face, $x_i \rightarrow 0$ and $u_i \rightarrow -\infty$ *with growing variance*: in Layer B’s own coordinate a taxon approaching the boundary is a *large-variance* direction — the exact opposite of the thin, collapsing direction it is geometrically. Estimating effective dimension by PCA on the u ’s would therefore *inflate* the dimension near faces and hide the very concentration we want to detect. The cube coordinate $\xi_i = x_i/x_k \in [0, 1]$ of Definition 2.1 behaves correctly: $x_i \rightarrow 0 \Rightarrow \xi_i \rightarrow 0$, a bounded variance collapse. The design already carries *both* coordinates, so the rule is clean and free.

Remark 8.3 (Estimate where the boundary is thin; model where dependence is unbounded). Detect the effective dimension and the soft face in the cube coordinate ξ (boundary = variance collapse); estimate the Layer B precision Ω in the log-ratio coordinate u (boundary = unbounded dependence, where the Gaussian / nonparanormal graphical model is appropriate). Doing dimension estimation in u inverts the sign of the effect.

In plain terms. The log scale stretches an “almost-absent” taxon out toward minus infinity, which makes it look like the *biggest* source of variation in the sample. On the raw cube scale the same taxon correctly looks tiny. So measure *how many directions matter* on the cube scale, but model the fine-grained dependence among the taxa that do matter on the log scale.

8.5 Use ℓ_1 , not ℓ_2 , to fuse the support layer

The synchronized-field objective (Definition 4.1) fuses all three blocks across charts with *squared* Frobenius terms, $\|\mu_c - \mu_{c'}\|_2^2 + \eta_A \|B_c - B_{c'}\|_F^2 + \eta_B \|\Omega_c - \Omega_{c'}\|_F^2$. The ℓ_2 choice is right for the *continuous* blocks μ and Ω , which vary smoothly within a stratum. It is the wrong choice for the *discrete support layer* (B , and the face/dimension itself), which is supposed to *reorganize sharply*

across a community-state transition. Indeed Experiment 4 (E4) states the thesis as “the field reorganizes across CST transitions,” with kill criterion “ $\mathcal{A}$ constant across the boundary” — yet an ℓ_2 fusion of B across a boundary face *enforces continuity* of the co-occurrence graph, working directly against the effect E4 is trying to detect. The fix is to match the penalty to the nature of the field:

- fuse the continuous blocks μ, Ω with ℓ_2 (smooth within a stratum, as now);
- fuse the discrete support blocks (presence π , edges B , and the effective dimension / soft face) with an ℓ_1 /total-variation penalty $\sum_{c \sim c'} \zeta_{cc'} \|B_c - B_{c'}\|_1$ (and likewise for the dimension field), which permits a clean switch at the boundary while still denoising within a stratum.

Operationally the effective-dimension field *gates the fusion graph*: a synchronization edge $c \sim c'$ that crosses a detected stratum boundary (a jump in k_c or a change in the soft face) is down-weighted or cut for the *discrete* layer while kept for the *continuous* layer. Sharp where the geometry is sharp; smooth where it is smooth.

In plain terms. Averaging two neighboring wiring diagrams blurs them together. That is fine for gradually changing abundance *levels*, but wrong for the on/off *wiring*, which is meant to flip at a community-state boundary. Use a smoothing penalty for the levels and a switch-tolerant penalty (the kind that allows a few sharp jumps and otherwise stays flat) for the wiring — and let the detected boundaries decide which neighboring charts are even allowed to be averaged together.

8.6 The stratified dimension field as an exactly-solvable graph problem

The effective-dimension field is a labeling of the sample graph by integers $k_c \in \{1, \dots, D_{\max}\}$. Choosing it jointly is a graph-regularized labeling problem,

$$\min_{\{k_c\}} \sum_c c_c(k_c; h_{\text{band}}) + \lambda \sum_{c \sim c'} \zeta_{cc'} \rho(k_c, k_{c'}),$$

with per-anchor cost c_c from Definition 8.1 (computed in ξ , at the bandwidth scale) and a pairwise penalty ρ . Two facts make this practical rather than heuristic.

1. **Convex TV is exactly minimizable.** For the total-variation penalty $\rho(a, b) = |a - b|$ on the linearly ordered label set $\{1, \dots, D_{\max}\}$, the global optimum is found in polynomial time by a single graph cut (Ishikawa’s construction [16]) — no continuous relaxation, no rounding, no local minima. TV penalizes the *size* of a dimension jump (a $1 \rightarrow 3$ jump costs more than $1 \rightarrow 2$), a sensible bias near a singularity. This is the same fused-lasso [17] idea used within charts, now on the integer dimension field.
2. **Strata have a partial order.** A faithful stratification obeys a frontier condition: the boundary of a k -stratum is a union of strata of dimension $< k$ (the closure of a 2-face contains its edges and vertices, not vice versa), so dimension is lower-semicontinuous and jumps are *directional*. A symmetric penalty (TV or the Potts penalty $\rho = \mathbf{1}\{a \neq b\}$, solved approximately by α -expansion [18]) ignores this; the faithful version encodes the simplex face lattice as a partial order on labels, for which convex-prior MRF optimization is also available [19]. This whole problem — recovering strata and their dimensions from a noisy point cloud near singularities — is the subject of *stratification learning* [21, 22].

In plain terms. Each sample proposes a dimension from its own neighborhood; the penalty makes neighbors agree unless there is a real reason to disagree (a stratum boundary). Because the thing being chosen is an ordered whole number, this tug-of-war has a fast, exact solution — you do

not have to guess and round. And because real strata fit together like the faces of a shape (edges bound triangles, not the reverse), the most faithful version also remembers which jumps are allowed to go which way.

8.7 A new read-out: present-and-active versus present-but-dormant

Separating the combinatorial face from the effective dimension produces an object neither layer names alone. A taxon can be *present* (an edge candidate in Layer A) yet contribute a *collapsed* direction (adding it does not raise k_c): it is *present-but-dormant* — hanging on near the detection limit without fluctuating. A taxon that is present *and* raises the effective dimension is *present-and-active*. This distinction — a marginal colonizer versus an engaged community member — is biologically meaningful and is invisible to Layer A (presence only) and to Layer B (dependence conditional on presence). It is a candidate read-out in its own right and a natural covariate for the host coupling of Section 3.3.

8.8 Honest scope and the detectability floor

Two cautions keep this from being oversold. *Where it pays.* In the vaginal communities the active sets are small (“a handful of co-present taxa”), so $|A| - 1$ is already 2–4 and the further reduction to k_c is often second-order. The within-face rank reduction earns its keep most where faces are large — higher-diversity gut communities and the metagenomic transfer (E8). For the minimal sprint the two high-value pieces do *not* depend on large faces: the *soft face* (so the presence threshold stops being a hard, read-depth-confounded cut) and the ℓ_1 *fusion of the support layer* (so synchronization stops fighting the E4 boundary- reorganization thesis). *Where it fails.* The detectability of a low-dimensional concentration is floored by measurement noise: in 16S the “thickness” of a near-face tube may *be* finite-read-depth multinomial noise rather than real structure, and from 10–20 local points one cannot cleanly separate “structurally k -dimensional concentration” from “ $|A| - 1$ -dimensional process observed with thin-looking noise.” The principled escapes are the multi-scale persistence of Section 8.3 and *replicates* (technical or biological), which estimate the noise floor and let one ask whether a thin direction survives above it. Absent those, “the dimension here is k ” is meaningful only with an attached scale and noise model.

8.9 Summary of changes to the model object

1. **Stratum.** Replace the single combinatorial face A by the refined stratum (A, k_c) (Def. 8.1); the face is the cap, k_c the working dimension.
2. **Layer B.** Estimate Ω on the k_c retained directions, not the full active set.
3. **Face.** Soften the presence threshold into a scale-coupled effective active set A_h (Def. 8.2), selected like β by held-out association likelihood.
4. **Coordinates.** Estimate dimension/soft-face in ξ , model dependence in u (Remark 8.3).
5. **Synchronization.** ℓ_2 fusion for μ, Ω ; ℓ_1 /TV fusion for the support layer (π, B , dimension), gated by the effective-dimension field (Section 8.5).
6. **Read-out.** Add present-and-active versus present-but-dormant (Section 8.7).

8.10 Experiment and kill criterion

Experiment 9 (Soft-face / effective-dimension refinement). On the two largest L^∞ -CSTs, compare three substrates under identical Layer-A and Layer-B estimators and identical folds: (i) the hard-

support face $A = \text{supp}(x)$ (baseline); (ii) the soft face A_h (Def. 8.2); (iii) the refined stratum (A_h, k_c) with Ω estimated at the effective rank (Def. 8.1). Score by held-out association-layer log-likelihood and by out-of-sample edge stability (stability selection frequency). Sweep the soft-face scale and the dimension bandwidth jointly with the LPS bandwidth. As a directed diagnostic, on a synthetic boundary-supported truth (a known tube of radius ρ around a k -face), verify that the effective-dimension estimate in ξ recovers k at scales $\gtrsim \rho$ while the same estimate in u inflates it (Remark 8.3); and verify that ℓ_1 support fusion preserves a planted edge reorganization across a boundary that ℓ_2 fusion smooths away.

Kill criterion. The refinement is falsified if, out-of-sample, the soft face and the refined-stratum substrates never beat the hard-support baseline on held-out association likelihood or edge stability, *and* the ξ -versus- u and ℓ_1 -versus- ℓ_2 diagnostics show no difference on the synthetic truth. In that case the combinatorial face is already sufficient and the extra machinery should be dropped.

References

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